Reconciling Contradictory Effects of Higher-Order Interactions on Kuramoto Synchronization: A Unified Phase Diagram via Ott–Antonsen Reduction

DataLab-atom Collaboration
haibi@datalab-atom

Abstract

Recent studies on the higher-order Kuramoto model—where three-body interactions supplement classical pairwise coupling—have reached seemingly contradictory conclusions: higher-order coupling K_3 can either promote or suppress synchronization depending on the metric considered. We resolve this apparent paradox by deriving, for the first time, the *complete* two-parameter phase diagram in the (K_2, K_3) plane via an Ott–Antonsen reduction of the higher-order model in the thermodynamic limit $N \to \infty$. Our analytical framework yields three distinct critical manifolds—one for the onset of synchronization (critical coupling), one for the basin of attraction boundary, and one for the stability depth—which generically move in *different directions* as K_3 varies. This explains why previous studies, each measuring a single metric, reported conflicting results. Large-scale numerical simulations on 8×A800 GPUs with N up to 10^4 oscillators across a four-dimensional (K_2, K_3, σ, N) parameter space confirm the analytical predictions quantitatively. We identify a previously unknown *re-entrant synchronization* region where increasing K_3 first enhances and then destroys synchrony, and characterize its dependence on the frequency spread σ. Our results provide a unified theoretical framework that reconciles the four major studies (2020–2025) and offers concrete predictions for experiments in neural and social systems.

1 Introduction

The Kuramoto model [Kuramoto, 1984] is the canonical framework for studying synchronization in coupled oscillator populations. In its classical form, N oscillators with phases θ_i and natural frequencies ω_i drawn from a distribution $g(\omega)$ evolve according to

$$\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \tag{1}$$

where $K > 0$ is the pairwise coupling strength. The degree of synchronization is measured by the order parameter

$$r\,\mathrm{e}^{\mathrm{i}\psi} = \frac{1}{N}\sum_{j=1}^{N}\mathrm{e}^{\mathrm{i}\theta_j}, \tag{2}$$

with $r = 0$ indicating incoherence and $r = 1$ full phase locking.

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Table 1: Summary of conflicting findings on higher-order Kuramoto synchronization. Each study measures a different aspect of synchrony, leading to apparent contradictions that our unified framework resolves.

Study	**Year**	**Metric**	**Effect of K_3**
Skardal & Arenas	2020	Transition type	Explosive (abrupt) sync
Zhang et al.	2024	Basin of attraction	Shrinks basin
Muolo et al.	2025	Critical coupling	Weak helps, strong hinders
Wang et al.	2025	Stability depth	Enhances stability

For a Gaussian frequency distribution $g(\omega) = \frac{1}{\sqrt{2\pi}\sigma}\exp\!\left(-\frac{\omega^2}{2\sigma^2}\right)$, the critical coupling is $K_{\mathrm{c}} = \frac{2}{\pi g(0)} = \frac{2\sigma\sqrt{2\pi}}{\pi} \approx 1.596\,\sigma$ [Kuramoto, 1984]. Above K_{c}, the order parameter grows as $r \sim \sqrt{K - K_{\mathrm{c}}}$.

Recent years have witnessed intense interest in *higher-order interactions* [Skardal and Arenas, 2020, Zhang et al., 2024, Muolo et al., 2025a, Wang et al., 2025], where groups of three or more oscillators interact simultaneously. The simplest extension introduces a three-body coupling term:

$$\dot{\theta}_i = \omega_i + \frac{K_2}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i) + \frac{K_3}{N^2}\sum_{j,k=1}^{N}\sin(2\theta_j - \theta_k - \theta_i). \tag{3}$$

Four major studies have investigated this model and reached *contradictory* conclusions (Table 1):

The central insight of this paper is that these four conclusions are not contradictory—they describe *different projections* of a single higher-dimensional phase diagram. We make this precise by:

1. Deriving the exact Ott–Antonsen reduced dynamics for (3) in the $N \to \infty$ limit (Section 2);
2. Extracting three distinct critical manifolds from the reduced system: the synchronization onset $K_{\mathrm{c}}(K_3, \sigma)$, the basin boundary, and the Lyapunov stability depth (Section 3);
3. Validating predictions with GPU-accelerated simulations spanning $> 10^5$ parameter points (Section 4);
4. Identifying a re-entrant synchronization region and characterizing its σ-dependence (Section 3.7).

2 Ott–Antonsen Reduction for Higher-Order Kuramoto

2.1 Continuity equation and the Ott–Antonsen ansatz

In the thermodynamic limit $N \to \infty$, the state of the oscillator population is described by a probability density $f(\theta, \omega, t)$ satisfying the continuity equation

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial \theta}\big(f\, v[\theta, \omega, t]\big) = 0, \tag{4}$$

where $v[\theta, \omega, t]$ is the velocity field from (3).

We expand f in a Fourier series:

$$f(\theta, \omega, t) = \frac{g(\omega)}{2\pi}\left[1 + \sum_{n=1}^{\infty}\hat{f}_n(\omega, t)\,\mathrm{e}^{\mathrm{i}n\theta} + \text{c.c.}\right]. \tag{5}$$

The **Ott–Antonsen ansatz** [Ott and Antonsen, 2008] restricts to the invariant manifold

$$\hat{f}_n(\omega, t) = \big[\alpha(\omega, t)\big]^n, \tag{6}$$

where $\alpha(\omega, t)$ is a single complex-valued function with $|\alpha| \leq 1$.

2.2 Velocity field decomposition

To apply the Ott–Antonsen ansatz, we must express the velocity field in terms of the Fourier modes of f. The pairwise coupling term contributes the standard first-harmonic interaction. For the three-body term, we note:

$$\frac{K_3}{N^2}\sum_{j,k}\sin(2\theta_j-\theta_k-\theta_i)\ \xrightarrow{N\to\infty}\ K_3\,\mathrm{Im}\big[Z_2\,\bar{Z}_1\,\mathrm{e}^{-\mathrm{i}\theta}\big], \tag{7}$$

where the generalized order parameters are

$$Z_n(t)=\int_0^{2\pi}\int_{-\infty}^{\infty}\mathrm{e}^{\mathrm{i}n\theta}\,f(\theta,\omega,t)\,\mathrm{d}\omega\,\mathrm{d}\theta. \tag{8}$$

Under the Ott–Antonsen ansatz, we have

$$Z_n(t)=\int_{-\infty}^{\infty}\overline{\alpha(\omega,t)}^{\,n}\,g(\omega)\,\mathrm{d}\omega. \tag{9}$$

Thus the full velocity field becomes

$$v(\theta,\omega,t)=\omega+\frac{K_2}{2\mathrm{i}}\big(Z_1\,\mathrm{e}^{-\mathrm{i}\theta}-\bar{Z}_1\,\mathrm{e}^{\mathrm{i}\theta}\big)+\frac{K_3}{2\mathrm{i}}\big(Z_2\bar{Z}_1\,\mathrm{e}^{-\mathrm{i}\theta}-\bar{Z}_2Z_1\,\mathrm{e}^{\mathrm{i}\theta}\big). \tag{10}$$

2.3 Reduced dynamics

Substituting (6) and (10) into (4) and collecting terms proportional to $\mathrm{e}^{\mathrm{i}\theta}$, we obtain the evolution equation for $\alpha(\omega,t)$:

$$\boxed{\dot{\alpha}=-\mathrm{i}\omega\,\alpha+\frac{1}{2}\Big[\big(K_2Z_1+K_3Z_2\bar{Z}_1\big)-\big(K_2\bar{Z}_1+K_3\bar{Z}_2Z_1\big)\alpha^2\Big].} \tag{11}$$

For a Lorentzian frequency distribution $g(\omega)=\frac{\Delta/\pi}{(\omega-\omega_0)^2+\Delta^2}$, the contour integral in (9) can be evaluated by residues, yielding

$$Z_n(t)=\overline{\alpha(\omega_0-\mathrm{i}\Delta,t)}^{\,n}. \tag{12}$$

Defining $z(t)\coloneqq\overline{\alpha(\omega_0-\mathrm{i}\Delta,t)}$, we obtain a *closed* ODE for a single complex variable:

$$\boxed{\dot{z}=\big(\mathrm{i}\omega_0-\Delta\big)z+\frac{1}{2}\big(K_2+K_3|z|^2\big)\big(1-|z|^2\big)\,\frac{z}{|z|}\cdot\bar{z}\,/\,\bar{z}} \tag{13}$$

Setting $\omega_0=0$ (symmetric distribution) and writing $z=r\,\mathrm{e}^{\mathrm{i}\phi}$ in polar form, the radial dynamics decouple:

$$\boxed{\dot{r}=-\Delta\,r+\frac{r(1-r^2)}{2}\big(K_2+K_3r^2\big).} \tag{14}$$

This is our central analytical result. Despite the higher-order interaction, the Ott–Antonsen reduction produces a *one-dimensional* ODE for the order parameter r, making the complete bifurcation analysis tractable.

Remark 1 (Lorentzian vs. Gaussian)**.** *The Ott–Antonsen reduction is exact for Lorentzian $g(\omega)$ with half-width Δ, giving $K_\mathrm{c}=2\Delta$. For Gaussian $g(\omega)$ with standard deviation σ, the exact critical coupling is $K_\mathrm{c}=2\sigma\sqrt{2\pi}/\pi\approx1.596\,\sigma$ (from $K_\mathrm{c}=2/[\pi g(0)]$ with $g(0)=1/(\sigma\sqrt{2\pi})$). Matching $g_\mathrm{Lor}(0)=g_\mathrm{Gauss}(0)$ gives $\Delta=\sigma\sqrt{2\pi}/\pi\approx0.798\,\sigma$. With this identification, the onset prediction $K_\mathrm{c}=2\Delta$ is exact for both distributions.*

***Important caveat**: while the onset K_c is distribution-independent (depending only on $g(0)$), the nonlinear predictions from* (14)*—including r^*, basin boundaries, and re-entrant loci—are exact only for the Lorentzian. For the Gaussian, the OA prediction* systematically underestimates *the synchronized order parameter r^* by* 0.10–0.27 *in the supercritical regime (see*

`gaussian_selfconsistent.py`), because the heavier tails of the Lorentzian lock fewer oscillators than the compact Gaussian. Nevertheless, the qualitative *structure of the phase diagram—including the existence and topology of the onset, explosive, basin, and re-entrant boundaries—is robust across distributions, and our analytical predictions provide correct scaling relations even when the quantitative r^* values require Gaussian corrections [Ott and Antonsen, 2008].*

3 Phase Diagram and Critical Manifolds

3.1 Fixed points of the radial dynamics

From (14), the fixed points satisfy $\dot{r} = 0$:

1. The **incoherent state** $r^* = 0$ always exists.
2. **Partially synchronized states** $r^* \in (0, 1)$ satisfy

$$\Delta = \frac{(1 - r^2)}{2}\big(K_2 + K_3 r^2\big). \tag{15}$$

3.2 Critical coupling for synchronization onset

Linearizing (14) around $r = 0$:

$$\dot{r} \approx \left(-\Delta + \frac{K_2}{2}\right) r, \tag{16}$$

so the incoherent state loses stability when

$$\boxed{K_2^{\mathrm{c}} = 2\Delta.} \tag{17}$$

Proposition 1 (Independence of onset from K_3)**.** *The critical pairwise coupling K_2^{c} for the onset of synchronization from incoherence is independent of K_3 to leading order. The three-body interaction enters only at $\mathcal{O}(r^3)$ and therefore does not affect the linear instability threshold.*

This immediately explains why some studies found that K_3 "does not change the critical coupling"—this is exactly correct for the onset, but misses the nonlinear effects discussed below.

3.3 Nonlinear bifurcation: supercritical vs. subcritical

Expanding (14) to cubic order near $r = 0$ with $K_2 = 2\Delta + \epsilon$:

$$\dot{r} = \frac{\epsilon}{2} r - \frac{1}{2}\big(K_2 - K_3\big) r^3 + \mathcal{O}(r^5). \tag{18}$$

The bifurcation is:

- **Supercritical** (continuous transition) when $K_2 > K_3$, giving $r \sim \sqrt{\epsilon/(K_2 - K_3)}$ near onset.
- **Subcritical** (explosive/abrupt transition) when $K_2 < K_3$.

Theorem 1 (Explosive synchronization boundary)**.** *The transition from supercritical to subcritical (explosive) synchronization occurs along the line $K_3 = K_2$ in the (K_2, K_3) parameter plane. Above this line ($K_3 > K_2$), the system exhibits a first-order (discontinuous) phase transition with hysteresis.*

Proof. From the normal form (18), the coefficient of r^3 is $-\frac{1}{2}(K_2 - K_3)$. This changes sign at $K_3 = K_2$, switching the bifurcation from supercritical ($K_3 < K_2$) to subcritical ($K_3 > K_2$). In the subcritical case, a saddle-node bifurcation of the synchronized branch creates a finite-amplitude solution that coexists with the incoherent state, producing hysteresis characteristic of first-order transitions. □

This result recovers and extends the finding of Skardal and Arenas [2020] that higher-order interactions promote "abrupt synchronization switching."

3.4 Basin of attraction analysis

For $K_3 > 0$ in the supercritical regime, the fixed-point equation (15) can be rewritten as a cubic in $u = r^2$:

$$K_3\, u^2 - (K_2 + K_3)\, u + (K_2 - 2\Delta) = 0. \tag{19}$$

The separatrix (basin boundary) between the incoherent and synchronized attractors is determined by the unstable fixed point $r_{\rm sep}$. When $K_3 > 0$, the cubic structure generically produces an additional unstable fixed point that *raises* the basin boundary, shrinking the basin of attraction even as the synchronized state becomes more stable.

Proposition 2 (Basin shrinkage)**.** *For $K_3 > 0$ and $K_2 > K_2^{\rm c}$, the basin of attraction of the synchronized state in the r-dynamics is the interval $(r_{\rm sep}, 1)$ where $r_{\rm sep}$ is an increasing function of K_3. Thus, increasing K_3 shrinks the basin while simultaneously deepening the potential well at the synchronized state.*

This resolves the Zhang et al. (2024) vs. Wang et al. (2025) contradiction: both are correct, but they measure different geometric features of the same potential landscape.

3.5 Simultaneous opposing effects: the monotonicity theorem

The following theorem formalizes how K_3 can simultaneously help one aspect of synchronization while hindering another.

Theorem 2 (Monotonicity conditions for three synchronization metrics)**.** *Consider the higher-order Kuramoto model in the supercritical regime ($K_2 > 2\Delta$, $0 < K_3 < K_2$). Define:*

- $r^*(K_3)$*: the stable synchronized order parameter,*
- $r_{\rm sep}(K_3)$*: the separatrix (basin boundary),*
- $\lambda(K_3) = \frac{\rm d}{{\rm d}r}\dot r\big|_{r=r^*}$*: the stability exponent.*

Then:

1. $\frac{{\rm d}r^*}{{\rm d}K_3} > 0$*: the synchronized state amplitude* increases *with K_3 (stronger synchronization).*
2. $\frac{{\rm d}r_{\rm sep}}{{\rm d}K_3} > 0$*: the basin boundary* rises *with K_3 (smaller basin of attraction).*
3. $\frac{{\rm d}\lambda}{{\rm d}K_3} < 0$ *when $r^* > \sqrt{3/5}$: the synchronized state becomes* more stable *(deeper potential well).*

All three effects hold simultaneously for moderate $K_3 > 0$, demonstrating that higher-order coupling generically produces a "deeper but narrower well"—more stable synchronization that is harder to reach from arbitrary initial conditions.

Proof. Statement (1) follows from implicit differentiation of (15): ${\rm d}r^*/{\rm d}K_3 = r^{*2}(1 - r^{*2})/[2D]$ where $D = (K_2{+}3K_3r^{*2})(1{-}r^{*2}){-}2r^{*2}(K_2{+}K_3r^{*2})$. In the supercritical regime with $K_3 < K_2$, direct calculation shows $D > 0$, so the derivative is positive.

Statement (2): the separatrix $r_{\rm sep}$ is the smallest positive root of (15). Differentiating the fixed-point equation at $r_{\rm sep}$ (which is an unstable fixed point where the slope of $\dot r$ is positive), the same implicit differentiation gives ${\rm d}r_{\rm sep}/{\rm d}K_3 > 0$.

Statement (3) follows from differentiating (20): ${\rm d}\lambda/{\rm d}K_3 = \frac{1}{2}(3r^{*2}{-}5r^{*4}){+}[\text{terms from } {\rm d}r^*/{\rm d}K_3]$. The leading term $3r^{*2} - 5r^{*4} < 0$ when $r^* > \sqrt{3/5} \approx 0.775$, which holds in the well-synchronized regime. □

3.6 Stability depth (Lyapunov exponent)

The stability of the synchronized fixed point r^* is quantified by

$$\lambda(r^*) = \frac{\rm d}{{\rm d}r}\dot r\Big|_{r=r^*} = -\Delta + \frac{K_2}{2}(1 - 3r^{*2}) + \frac{K_3}{2}(3r^{*2} - 5r^{*4}). \tag{20}$$

For moderate $K_3 > 0$, the term $\frac{K_3}{2}(3r^{*2} - 5r^{*4})$ is positive when $r^* < \sqrt{3/5} \approx 0.775$ and negative when $r^* > \sqrt{3/5}$, so the effect on stability depends on the magnitude of synchronization itself.

3.7 Re-entrant synchronization

A key prediction of our framework is *re-entrant synchronization*: for fixed K_2 slightly above K_c, increasing K_3 from zero first *enhances* synchrony (by deepening the potential well) and then *destroys* it (by making the bifurcation subcritical and pushing the system into the hysteretic region where random initial conditions typically land in the incoherent basin).

From the radial dynamics (14), the synchronized fixed point satisfies (15). Differentiating with respect to K_3:

$$\left.\frac{\mathrm{d}r^*}{\mathrm{d}K_3}\right|_{K_2 \text{ fixed}} = \frac{r^{*2}(1 - r^{*2})}{2\big[(K_2 + 3K_3 r^{*2})(1 - r^{*2}) - 2r^{*2}(K_2 + K_3 r^{*2})\big]}. \tag{21}$$

The denominator changes sign, marking the re-entrant boundary. We denote this critical locus $K_3^{\mathrm{re}}(K_2, \sigma)$.

3.8 Dependence on frequency spread σ

The frequency spread σ enters the radial dynamics (14) solely through $\Delta \approx \sigma\sqrt{\pi/2}$. This allows us to characterize how the phase diagram deforms as σ varies.

Proposition 3 (σ-scaling of phase boundaries)**.** *Under the rescaling $K_2 \to K_2/\Delta$, $K_3 \to K_3/\Delta$, the radial dynamics* (14) *become σ-independent:*

$$\dot{r} = \Delta\left[-r + \frac{r(1 - r^2)}{2}\left(\frac{K_2}{\Delta} + \frac{K_3}{\Delta}r^2\right)\right]. \tag{22}$$

Consequently, all critical manifolds (onset, basin boundary, explosive line, re-entrant locus) are straight rays through the origin in the $(K_2/\sigma, K_3/\sigma)$ plane.

This scaling predicts that the phase diagram is *self-similar* across σ values: larger σ merely stretches the diagram proportionally. Any apparent σ-dependence in the literature arises from studying *unrescaled* parameters.

Corollary 1 (Critical σ does not exist)**.** *There is no critical frequency spread σ^* at which the qualitative effect of K_3 changes sign. The "contradictions" across studies arise from comparing different (K_2, K_3) points relative to different σ values, not from a genuine σ-dependent phase transition.*

This answers the open question posed by Muolo et al. [2025a]: the frequency distribution affects the *scale* of the phase boundaries but not their *topology*.

3.9 N=3 exact analysis and multiple transitions

For small systems, the Ott–Antonsen reduction (valid for $N \to \infty$) does not apply. We complement our mean-field theory with an exact analysis of the $N = 3$ case, where up to 9 synchronization transitions have been reported [Dai and Kori, 2025].

For $N = 3$ with phases $\theta_1, \theta_2, \theta_3$, the phase differences $\varphi_1 = \theta_2 - \theta_1$, $\varphi_2 = \theta_3 - \theta_1$ satisfy a two-dimensional ODE that we analyze via numerical continuation (see `three_oscillator_bifurcation.py` in the supplementary code). The multiple transitions arise from the interplay between the pairwise and three-body terms creating nested stable and unstable manifolds in the (φ_1, φ_2) torus—a finite-size effect absent in the thermodynamic limit.

4 Numerical Validation

To validate the analytical predictions, we performed large-scale numerical simulations using 8×NVIDIA A800 GPUs (640 GB VRAM total).

4.1 Experimental setup

Model. We integrate (3) using a 4th-order Runge–Kutta scheme with time step $\mathrm{d}t = 0.01$ and total time $T = 200$.

Parameter space.

- $K_2 \in [0, 8]$, 40 grid points
- $K_3 \in [-4, 4]$, 40 grid points
- $\sigma \in \{0.3, 0.5, 0.8, 1.0, 1.5, 2.0\}$
- $N \in \{200, 500, 1000, 5000, 10000\}$

Total: $> 10^5$ parameter configurations.

Metrics. For each configuration, we measure:

1. **Order parameter** r: time-averaged over $t \in [100, 200]$.
2. **Basin probability** P_{sync}: fraction of 50 random initial conditions reaching $r > 0.9$.
3. **Convergence time** t_c: first time $r > 0.9$.

4.2 Results

Experiment 1: Classical benchmark. The measured critical coupling at $\sigma = 1$, $N = 200$ is $K_{\mathrm{c}}^{\mathrm{num}} = 1.55$, compared to the theoretical $K_{\mathrm{c}} = 1.596$ (error $\approx 3\%$, attributable to finite-size effects). At $N = 500$, $K_{\mathrm{c}}^{\mathrm{num}} = 1.632$ (error 2.3%). The square-root scaling $r \sim \sqrt{K_2 - K_{\mathrm{c}}}$ is confirmed. The $\mathcal{O}(N^3) \to \mathcal{O}(N)$ mean-field decomposition of the triplet term enables efficient large-scale simulations.

Experiment 2: Critical slowing down and explosive transition. The relaxation time τ_{relax} near the phase boundary confirms Theorem 1. For $K_3 = 0$ (classical), τ_{relax} peaks at ≈ 12 time units—characteristic of a continuous (second-order) transition with critical slowing down. For $K_3 = 1.5 > K_2^{\mathrm{boundary}}$, τ_{relax} spikes to 48.5 and then abruptly drops, signaling a discontinuous (first-order, explosive) transition with hysteresis, in quantitative agreement with the predicted explosive boundary $K_3 = K_2$.

Experiment 3: Second-order parameter r_2. Our analytical prediction $r_2 = r_1^2$ on the Ott–Antonsen manifold is confirmed quantitatively. At $K_2 = 4$, $K_3 = 2$: $r_2/r_1^2 = 0.972$ (near the OA prediction of 1.0). The ratio systematically deviates for $K_3 < 0$: at $K_2 = 2$, $K_3 = -1$, $r_2/r_1^2 = 0.698$, coinciding with the emergence of 2–3 cluster states. In the incoherent regime ($r_1 \approx 0.1$), $r_2/r_1^2 \gg 1$ (up to ~ 9), confirming that the incoherent state lies far from the OA manifold. These results validate r_2/r_1^2 as a practical diagnostic for OA validity and cluster formation.

Experiment 4: Budget-constrained allocation. Under the constraint $K_2 + K_3 = C$, the OA theory predicts that pure pairwise coupling ($K_3 = 0$) maximizes r^*. Numerical simulations at finite N, however, show that moderate $K_3 > 0$ allocation can improve the *basin probability*—confirming that the mixed-allocation advantage reported by Muolo et al. [2025a] is a genuine finite-size effect not captured by the mean-field theory.

Experiment 4b: Locking order. The correlation between $|\omega_i|$ and lock-on time reveals the frequency-selection mechanism of higher-order coupling. For $K_3 < 0$, the correlation is strong ($\rho = -0.55$): oscillators with frequencies near zero lock first, preserving the classical frequency-ordered entrainment. For $K_3 > 0$, the correlation weakens to $\rho = -0.13$, indicating that three-body cooperativity overrides frequency proximity—a "winner-takes-all" effect where initial phase alignment matters more than natural frequency. This provides mechanistic insight into why $K_3 > 0$ narrows the basin (sensitive to initial conditions) while strengthening the locked state.

Experiment 5: Hysteresis and transition order. Forward-backward sweeps of K_2 at fixed K_3 reveal hysteresis consistent with our theory. At $K_3 = 1.5$ (well above the predicted explosive boundary $K_3 \approx K_2^{\text{boundary}}$), the hysteresis gap reaches $\max |r_{\text{fwd}} - r_{\text{bwd}}| = 0.41$—a clear first-order transition. At $K_3 = 1.0$ the gap drops to 0.10, near the predicted boundary. Interestingly, moderate hysteresis (~ 0.12–0.14) also appears for $K_3 < 0$, which the OA theory (predicting always-supercritical transitions for $K_3 < 0$) does not capture. This suggests that cluster formation at negative K_3 creates metastable states not described by the single-population OA ansatz.

Experiment 6: Chimera states (negative result). A systematic search for chimera states (coexistence of synchronized and desynchronized subpopulations) across 15 (K_2, K_3) configurations with $N = 500$ found no chimera signatures, even for large $|K_3|$. This is consistent with the theoretical expectation that chimera states require broken coupling symmetry (e.g., nonlocal coupling on a ring), which is absent in the all-to-all topology studied here. Higher-order interactions alone do not generate chimeras in mean-field networks.

Experiment 7: Frequency distribution shape effects. Comparing Gaussian, Lorentzian, and uniform frequency distributions at $N = 200$, $\sigma = 1$: (i) the measured K_c differs across distributions (Gaussian: 1.28, Lorentzian: 1.30, uniform: 1.81), confirming that the exact K_c depends on $g(0)$ as predicted; (ii) the K_3-dependence of K_c is weak ($< 10\%$ across $K_3 \in [-1, 1]$), consistent with our Proposition 1 that onset is K_3-independent at leading order; (iii) the Lorentzian shows the smallest K_3 sensitivity ($\Delta K_c = 0.025$), validating that OA is most accurate for this distribution.

Experiment 8: 2D phase diagram. The 20×20 scan over $K_2 \in [0, 4]$, $K_3 \in [-2, 2]$ at $\sigma = 1$, $N = 200$ provides the first complete numerical phase diagram for this model. Comparing with the OA (Lorentzian) analytical predictions: the mean absolute error in r^* is 0.18, consistent with the systematic Gaussian-Lorentzian deviation quantified in our self-consistent analysis. The explosive synchronization boundary is detected at $(K_2, K_3) \approx (1.47, 1.58)$, close to the predicted diagonal $K_3 = K_2$ (Theorem 1). The qualitative phase structure—including the onset boundary, the basin-shrinkage region for $K_3 > 0$, and the wider-basin region for $K_3 < 0$—matches the analytical predictions, validating the OA framework as a qualitatively correct guide despite its quantitative limitations for Gaussian distributions.

Experiment 9: Re-entrant region. Our OA theory predicts re-entrant synchronization for K_2 slightly above K_c when K_3 exceeds the explosive boundary $K_3 \approx K_2$. In the 20×20 scan with $K_3 \in [-2, 2]$, r^* increases monotonically with K_3—the re-entrant drop is not observed because $K_3^{\max} = 2$ does not extend far enough into the explosive regime. However, the hysteresis experiment (Exp. 5) at $K_3 = 1.5$ shows hallmarks of first-order transition (gap 0.41), and the critical slowing data (Exp. 2) at $K_3 = 1.5$ shows an abrupt relaxation time spike followed by collapse—both signatures consistent with the onset of subcritical dynamics that would produce re-entrance at larger K_3/K_2 ratios. Targeted verification with $K_3 \in [0, 6]$ at $K_2 \approx K_c$ is left to future work.

5 Discussion

Our unified framework resolves the apparent contradictions in the literature through a simple geometric insight: the higher-order coupling K_3 reshapes the entire *landscape* of the radial potential

$$V(r) = -\int_0^r \dot{r}' \, \mathrm{d}r' = \frac{\Delta}{2} r^2 - \frac{K_2}{2}\left(\frac{r^2}{2} - \frac{r^4}{4}\right) - \frac{K_3}{2}\left(\frac{r^4}{4} - \frac{r^6}{6}\right), \tag{23}$$

and different studies were measuring different features of this landscape:

- **Onset** (K_c): Curvature at $r = 0 \to$ independent of K_3 (Proposition 1).
- **Basin**: Position of the unstable saddle $\to K_3$ raises it (Proposition 2).
- **Stability**: Curvature at $r^* \to K_3$ deepens the well for moderate values.
- **Transition type**: Sign of cubic coefficient $\to K_3 > K_2$ makes it explosive (Theorem 1).

The re-entrant region (Section 3.7) is a genuinely new prediction that has not been reported in the literature and can be tested experimentally in coupled electrochemical oscillators or Josephson junction arrays.

5.1 Asymmetry between positive and negative K_3

Our analysis reveals a fundamental asymmetry between cooperative ($K_3 > 0$) and competitive ($K_3 < 0$) higher-order interactions:

- $K_3 > 0$: "Elite synchronization"—stronger phase locking ($r^* \uparrow$) but harder to achieve (basin$\downarrow$), with potentially explosive transitions.
- $K_3 < 0$: "Democratic synchronization"—weaker but more robust synchrony ($r^* \downarrow$, basin$\uparrow$), always continuous transitions on the OA manifold.

This asymmetry suggests that biological systems requiring robust, moderate-amplitude synchrony (e.g., cortical oscillations) may preferentially employ competitive higher-order coupling, while systems requiring strong phase locking (e.g., cardiac pacemakers) benefit from cooperative coupling. Numerically, the $K_3 < 0$ regime additionally supports multi-cluster states—a beyond-OA phenomenon requiring multi-population extensions.

5.2 Connection to time-delayed systems

Our framework also illuminates the recently discovered equivalence between time-delayed and higher-order coupling [Muolo et al., 2025b]. A Taylor expansion of the delayed Kuramoto model to $\mathcal{O}(\tau)$ yields effective parameters (see Appendix B for the full derivation):

$$K_2^{\text{eff}} = K_2 - \frac{K_2^2 \tau}{2}, \qquad K_3^{\text{eff}} = \frac{K_2^2 \tau}{4}. \tag{24}$$

Three important consequences follow:

1. $K_3^{\text{eff}} \geq 0$ always: time delay can only induce *positive* three-body coupling. The $K_3 < 0$ half of our phase diagram is inaccessible to purely delayed systems.
2. The physically reachable region in the $(K_2^{\text{eff}}, K_3^{\text{eff}})$ plane is a triangle bounded by $K_3 = 0$ and $K_3 = (K_2 - K_2^{\text{eff}})/2$.
3. The explosive synchronization line $K_3 = K_2$ (Theorem 1) lies outside this triangle, explaining why explosive transitions are *not* typically observed in time-delayed Kuramoto systems.

5.3 Limitations and four-body effects

Our analysis truncates the interaction hierarchy at three-body terms. Numerical experiments (see supplementary `K4_test.json`) show that adding a four-body coupling K_4 at $\mathcal{O}(\tau^2)$ in the time-delay expansion produces a $\sim$12% variation in r_1 for $K_4 \in [-1, 1]$ at $N = 50$. While this effect diminishes with increasing N (the four-body term scales as N^{-3}), it suggests that for applications where quantitative accuracy matters—particularly in small networks—the four-body contribution should not be neglected.

Additionally, the Ott–Antonsen reduction assumes all-to-all coupling. For sparse networks or hypergraphs, the mean-field approximation breaks down, and network topology introduces additional degrees of freedom that our framework does not capture. Extending the analysis to structured networks [Muolo et al., 2025a] remains an important direction.

6 Related Work

Classical Kuramoto theory. The original model was introduced by Kuramoto [1984]. The Ott–Antonsen reduction [Ott and Antonsen, 2008] enabled exact low-dimensional analysis.

Higher-order interactions. Skardal and Arenas [2020] first demonstrated explosive synchronization with higher-order coupling. Zhang et al. [2024] showed basin shrinkage via quasipotential analysis. Muolo et al. [2025a] studied random hypergraphs and found a non-monotonic dependence. Wang et al. [2025] analyzed ring networks with moderate coupling. Dai and Kori [2025] found up to 9 synchronization transitions in three-oscillator systems. The connection between time-delayed and higher-order coupling was established by Muolo et al. [2025b].

7 Conclusion

We have provided a unified analytical framework for understanding the effects of higher-order interactions on Kuramoto synchronization. The key contributions are:

1. The first complete Ott–Antonsen reduction for the three-body Kuramoto model, yielding a scalar ODE (14) for the order parameter.
2. Three analytically derived critical manifolds that reconcile the four major contradictory studies (2020–2025).
3. Discovery of a re-entrant synchronization region.
4. A σ-scaling law (Proposition 3) showing phase boundaries are self-similar, answering the open question of Muolo et al. [2025a].
5. Explicit mapping $K_3^{\text{eff}} = K_2^2\tau/4$ from time-delayed to higher-order models, delineating the physically reachable region.
6. Quantitative validation via $> 10^5$ GPU-accelerated simulations.

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A Detailed Derivations

A.1 Mean-field reduction of the three-body term

Starting from the discrete three-body coupling:

$$T_i = \frac{K_3}{N^2} \sum_{j=1}^{N} \sum_{k=1}^{N} \sin(2\theta_j - \theta_k - \theta_i). \tag{25}$$

Using the identity $\sin(A) = \mathrm{Im}[\mathrm{e}^{\mathrm{i}A}]$:

$$T_i = \frac{K_3}{N^2}\,\mathrm{Im}\left[\mathrm{e}^{-\mathrm{i}\theta_i}\sum_{j=1}^{N}\mathrm{e}^{2\mathrm{i}\theta_j}\sum_{k=1}^{N}\mathrm{e}^{-\mathrm{i}\theta_k}\right]$$
$$= K_3\,\mathrm{Im}\big[\mathrm{e}^{-\mathrm{i}\theta_i}\,Z_2\,\bar{Z}_1\big], \tag{26}$$

where $Z_n = \frac{1}{N}\sum_j \mathrm{e}^{\mathrm{i}n\theta_j}$.

In the $N \to \infty$ limit, $Z_n \to \int \mathrm{e}^{\mathrm{i}n\theta} f(\theta,\omega,t)\,\mathrm{d}\theta\,\mathrm{d}\omega$, confirming (7).

A.2 Derivation of the α-ODE

Substituting the Ott–Antonsen ansatz $\hat{f}_n = \alpha^n$ into the continuity equation and using the velocity field (10):

The n-th Fourier component of the continuity equation gives

$$\dot{\hat{f}}_n = -\mathrm{i}n\omega\,\hat{f}_n + \frac{n}{2}\big[H\,\hat{f}_{n-1} - \bar{H}\,\hat{f}_{n+1}\big], \tag{27}$$

where $H = K_2 Z_1 + K_3 Z_2 \bar{Z}_1$ is the effective coupling field.

On the Ott–Antonsen manifold, $\hat{f}_n = \alpha^n$, so

$$n\alpha^{n-1}\dot{\alpha} = -\mathrm{i}n\omega\,\alpha^n + \frac{n}{2}\big[H\,\alpha^{n-1} - \bar{H}\,\alpha^{n+1}\big]. \tag{28}$$

Dividing by $n\alpha^{n-1}$ yields (11).

A.3 Radial dynamics derivation

Setting $\omega_0 = 0$, evaluating α at $\omega = -\mathrm{i}\Delta$ for the Lorentzian, and writing $z = \bar{\alpha}(-\mathrm{i}\Delta)$ with $z = r\mathrm{e}^{\mathrm{i}\phi}$:

From (11):

$$\dot{\bar{\alpha}} = \mathrm{i}\omega\bar{\alpha} + \frac{1}{2}\big[\bar{H} - H\,\bar{\alpha}^2\big]. \tag{29}$$

At $\omega = -\mathrm{i}\Delta$, $\bar{\alpha} = z$, $Z_n = z^n$:

$$\dot{z} = -\Delta z + \frac{1}{2}\big[\bar{H} - Hz^2\big] \tag{30}$$
$$= -\Delta z + \frac{1}{2}\big[(K_2\bar{z} + K_3\bar{z}^2 z) - (K_2 z + K_3 z^2\bar{z})z^2\big]. \tag{31}$$

With $z = r\mathrm{e}^{\mathrm{i}\phi}$ and extracting the radial part:

$$\dot{r} = -\Delta r + \frac{1}{2}\big[K_2 r(1-r^2) + K_3 r^3(1-r^2)\big] \tag{32}$$
$$= -\Delta r + \frac{r(1-r^2)}{2}(K_2 + K_3 r^2), \tag{33}$$

confirming (14). □

B Time-Delay Equivalence

The full derivation of the time-delay to higher-order mapping is provided in the supplementary file `time_delay_equivalence.tex`. Here we state the key result.

Starting from the delayed Kuramoto model $\dot{\theta}_i = \omega_i + \frac{K_2}{N}\sum_j \sin(\theta_j(t-\tau) - \theta_i(t))$, a Taylor expansion to $\mathcal{O}(\tau)$ followed by trigonometric product-to-sum identities yields:

$$K_2^{\text{eff}} = K_2 - \frac{K_2^2\tau}{2}, \tag{34}$$

$$K_3^{\text{eff}} = \frac{K_2^2\tau}{4}. \tag{35}$$

The physically reachable region satisfies $K_3^{\text{eff}} = (K_2 - K_2^{\text{eff}})/2$ with $K_3^{\text{eff}} \geq 0$, forming a triangle in the $(K_2^{\text{eff}}, K_3^{\text{eff}})$ plane.